

# s1.05 — Internal metrics ingestion service

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## Interview framing

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- Target: OCI IC3 ownership of a bounded platform component.
  - `IC3 baseline` is what must land in every phase.
  - `IC4 stretch` adds architecture changes under stronger constraints.
  - These are preparation labels, not Oracle's official rubric.
- Evidence: retain the first-hand report attribution in [the bank](#).
  - The architecture below is a proposed interview answer.
  - It does not claim to reproduce Oracle's implementation or an interviewer's accepted answer.
- Why this is hard
  - Naive: every `host` posts samples straight to the `time-series database` and `retries on failure`.
  - First break: a database slowdown backs up into `100,000` hosts, retries double-count counters, and an in-memory queue acknowledgment loses points on crash.
  - Second break: a consumer crashes after writing but before committing progress, rollups double-add, averaged percentiles lie, and one tenant's label explosion evicts everyone's index.
  - Minimum primitives
    - A swap-buffer agent
    - A partitioned durable log acknowledged after quorum
    - Stable point identity
    - An atomic dedup-plus-update sink boundary
    - A `per-tenant cardinality` budget.
- 🗣️: "Acknowledge at the durable log, not the database; then make every downstream write idempotent so replay costs nothing but time."
- Opening script
  - "I will separate durable ingestion from query freshness and show how batching, identity, and bounded recovery fit together."
  - "First I confirm metric types, the acceptable loss policy, and how stale a dashboard may be."
  - "Then I size the log, the catch-up rate, and the cardinality budget from one million points per second."
  - "The deep dive is the agent flush race and the replay-safe sink boundary."
  - "I will close with the outage drill, tenant isolation, and where I would change the design if the data became billing-grade."
- Numerical targets below are interview assumptions unless a reference explicitly supplies them.

- Validate performance targets with the actual payload and deployment topology.

| Phase                      | Clock       | Deliverable                                              |
|----------------------------|-------------|----------------------------------------------------------|
| Intro and behavioral slice | 00:00–05:00 | 90-second pitch; confirm the prompt                      |
| Requirements               | 05:00–11:00 | Metric semantics, loss policy, primary invariant         |
| Estimates and contract     | 11:00–17:00 | Points/s, backlog math, batch API semantics              |
| Architecture               | 17:00–30:00 | Diagram, log/consumer/query ownership, one request trace |
| Deep dive                  | 30:00–45:00 | Agent flush race, point identity, replay-safe sink       |
| Operations and tradeoffs   | 45:00–50:00 | Failure table, outage drill, tradeoff, wrap script       |
| Wrap-up                    | 50:00–54:00 | Four-sentence recap and weakest assumption               |
| Questions for interviewer  | 54:00–60:00 | Two role-specific questions                              |

- Clock note
  - If the interviewer states 45 minutes, drop the intro to 2 minutes and compress each phase proportionally.
  - The interviewer's clock wins over this table.

## Requirements — IC3 baseline

- Numerical targets are interview assumptions unless an adjacent primary source states otherwise.

| Question                              | Assumed answer                                                    | Why it changes the design                                                      |
|---------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Which metric types?                   | Counters, gauges, histograms                                      | Each type needs its own <b>aggregation</b> ; one count/sum/max record is wrong |
| <b>Loss policy</b> before acceptance? | Operational telemetry; bounded counted drops allowed at the agent | Permits drop-on-full spool; billing-grade data would forbid it                 |
| Durability versus freshness priority? | Accepted data must survive sink outages; freshness may degrade    | Forces a durable log before any sink                                           |
| Query workload?                       | Dashboards over recent bounded ranges; no full-text log search    | Rollups plus raw; series fan-out caps                                          |
| Multi-tenant?                         | Yes; internal teams as tenants mapped to compartments             | Tenant sits in every series key, budget, and query path                        |
| <b>Out-of-order tolerance?</b>        | Bounded, about 5 min                                              | Lateness window, provisional rollups, versioned repair                         |
| Who owns the agent?                   | The <b>platform</b> ships the agent on every host                 | Producer incarnation and batch identity are ours to define                     |

- Functional scope
  - Accept **batched points** from an agent on every host.
  - Clarify the **metric** semantics.
    - Counters represent accumulated events and may reset on process restart.
    - Gauges represent sampled current values.
    - Histograms represent distributions through mergeable bucket counts.
    - Do not aggregate every metric into the same count/sum/max record.
  - Persist accepted points durably before any **sink** applies them.
  - Serve bounded dashboard queries over raw points and rollups.
  - Support authorized replay of a source range into a sink.
- Assumed scope
  - 100,000 hosts.
  - 100 active series/host.
  - One point per **series** every 10 s.
  - Dashboard queries over recent **bounded time ranges**.
  - Full-text log search is separate.

| Requirement                     | Target                                           | Enforced by                                                                                             |
|---------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Sustained ingest                | 1,000,000 points/s                               | Partitioned log sized with headroom; per-tenant admission budget                                        |
| Ingestion availability          | 99.95%                                           | Stateless gateway fleet across fault domains; replicated log quorum                                     |
| Batch acknowledgment latency    | p99 < 250 ms after gateway receipt               | Acknowledge after log quorum only; no synchronous database write                                        |
| Accepted-data freshness         | p99 < 30 s                                       | Consumer lag alarm; catch-up capacity of 1.5× live rate                                                 |
| Bounded dashboard query         | p99 < 2 s                                        | Series fan-out and point caps; rollups; query budget per tenant                                         |
| Replay log retention            | 24 h                                             | Retention configured above outage plus catch-up plus margin                                             |
| Lateness window                 | 5 min (assumption)                               | Provisional rollups until the window closes                                                             |
| Per-tenant active-series budget | 1 million (assumption)                           | Admission check at the gateway before a new series is created                                           |
| Control-plane independence      | 30 min outage drill with valid admission budgets | Gateways retain policy only within its declared 30 min lease; expiration invokes the stream fail policy |

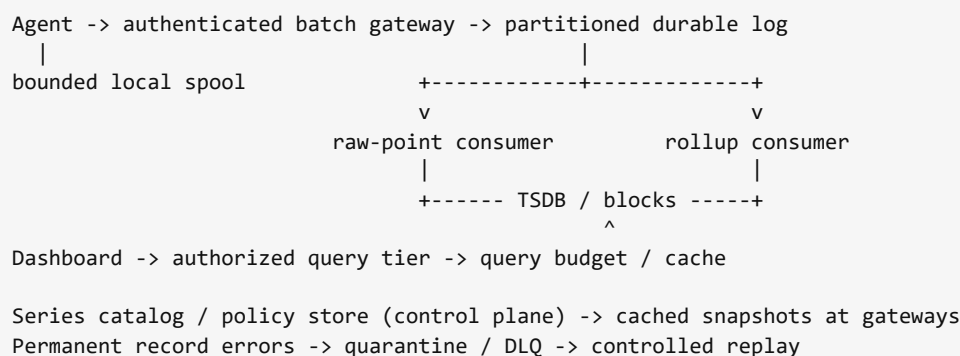
- Invariants
  - Acknowledged acceptance means the batch reached the chosen durable boundary.
  - A consumer retry must not double-add an already applied aggregate.
  - Tenant identity is part of every series and access path.
  - Queue and cardinality bounds are explicit.
  - Data before durable acceptance may be dropped only under the documented metrics policy.
- Non-goals
  - Billing or metering records with a no-loss contract.
  - Full-text log search or trace storage.
  - Cross-tenant metric federation.

## Capacity and consequence — IC3 baseline

- Input
  - $100,000 \times 100 / 10 = 1,000,000$  points/s.
  - At an assumed compact 50 B/point, raw input is 50 MB/s.
  - Raw daily volume is  $50 \text{ MB/s} \times 86,400 \approx 4.32 \text{ TB/day}$ .
  - Three replicated copies imply approximately 12.96 TB for a day before indexes and overhead.
  - Compression is workload-dependent and must be measured.
  - Decision: the log must be partitioned and retention-bounded
    - 24 h of replicated raw log is on the order of 13 TB, so retention is a capacity line item, not a default.
- Batching
  - One flush per host every 10 s gives 10,000 batch requests/s.
  - In this sampling scenario, batching reduces requests, not the one million logical points.
  - Local pre-aggregation reduces points only when the original measurements and query semantics allow it.
  - Decision: the gateway is sized for 10,000 requests/s of about 5 KB each
    - The log and sinks are sized for one million points, and no one may quote the smaller number for the larger system.
- Recovery capacity
  - A 2 h sink outage creates 7.2 billion points of backlog.
  - A recovery rate of 1.5 million/s leaves 0.5 million/s for catch-up.
  - Catch-up takes  $14,400 \text{ s} = 4 \text{ h}$  while live traffic continues.
  - The oldest affected record waits about six hours from outage start to full recovery.

- Retention must cover outage plus catch-up plus operational margin.
- Decision: provision sink write capacity at 1.5x live rate and set log retention to 24 h
  - A 2 h outage then consumes 6 h of the retention window and leaves margin.
- Series cardinality
  - 10 million active series can dominate TSDB memory and indexes.
  - Bound label combinations before accepting a new series.
  - Decision: admission control on new series is a gateway function with a per-tenant budget, because index memory is shared and cannot be reclaimed quickly.

## Architecture and flow — IC3 baseline



- Agent
    - Buys concurrent-safe batching through an atomic buffer swap and a bounded spool.
    - Cost: bounded local loss before acceptance under the documented drop policy.
  - Gateway
    - Validates schema and payload bounds before durable acceptance.
    - Applies tenant admission budgets.
    - Acknowledges only after the log reports the required durability condition.
    - Buys a single admission point for cardinality and tenancy.
    - Cost: one extra hop and the 250 ms acknowledgment budget must include the log quorum write.
  - Durable log
    - Decouples short storage outages from ingestion.
    - Supplies ordered partitions and replay.
    - It has finite retention and capacity.
    - Cost: retention bytes, partition management, and a second system to operate.
  - Consumers
    - Raw-point and rollup consumers apply idempotent writes and then advance recoverable progress.
    - Buys replay safety and independent scaling of raw versus rollup paths.
    - Cost: at-least-once delivery, so every sink write must carry a unique application identity.
  - Query tier
    - Uses independent workers and query budgets.
    - Separate processes do not automatically isolate shared disk, memory, or storage bandwidth.
    - Reserve storage capacity or use isolated read resources for expensive queries.
    - Cost: bounded results and freshness metadata replace the illusion of complete, current data.
  - Control plane versus data plane
    - Series catalog, tenant budgets, and retention policies live in the control plane.
    - Gateways and consumers hold cached snapshots so acceptance continues when the catalog is unavailable.
    - Cost: a budget change propagates with snapshot delay rather than instantly.
- Atomically swap the agent's active buffer for a fresh buffer.
- Serialize the detached batch outside its short lock.

- Send a stable batch identity with retry and jitter.
- Validate and durably append at the gateway.
- Acknowledge acceptance and release the local retry copy.
- Consumers apply idempotent writes before advancing recoverable progress.
- Queries read committed raw points or compatible rollups with freshness metadata.

## API and data model — IC3 baseline

| Surface                                   | Contract                                           | Conflict/error behavior                                                                                                              |
|-------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| POST /v1/metric-batches                   | Batch ID, producer incarnation, sequence, points   | 202 after durable append; same batch ID returns the prior result; partial acceptance is reported explicitly or the whole batch fails |
| GET /v1/metric-batches/{producer}/{batch} | Acceptance and processing outcome within retention | 404 after retention; authoritative answer after a lost 202                                                                           |
| GET /v1/query                             | Tenant-scoped selector, range, resolution          | 422 over fan-out or point caps; response carries freshness watermark                                                                 |
| GET /v1/series                            | Paginated names/labels                             | Prevent arbitrary cross-tenant enumeration; cursor scoped to tenant                                                                  |
| POST /v1/replays                          | Restricted source range and transform version      | Idempotency key, rate limit, and audit; 409 on overlapping active replay                                                             |

| State               | Identity/key                                        | Invariant                                                               |
|---------------------|-----------------------------------------------------|-------------------------------------------------------------------------|
| Series              | (tenant, canonical name, sorted labels)             | Full canonical identity resolves hash collisions                        |
| Logical point       | (series, producer incarnation, sample sequence)     | Retry keeps identity; source timestamp alone may collide                |
| Log record          | (partition, offset)                                 | Recovery coordinate for this append, not cross-replay business identity |
| Rollup contribution | (window, series, source batch ID)                   | Unique application boundary prevents double addition                    |
| Materialized window | (tenant, series, window start, resolution, version) | Rebuild can replace a complete version atomically                       |

- Reference DDL
  - Keys and uniqueness match the state table above and the [companion schema](#).
  - Logical points and log records live in the partitioned log and the TSDB; SQL holds the series catalog, batch outcomes, rollup contributions and windows, and replay records.

```

CREATE TABLE metric_series (
    tenant_id text NOT NULL, series_id text NOT NULL,          -- hash of the canonical identity; never
    trusted alone
    name text NOT NULL, canonical_labels text NOT NULL,        -- sorted label pairs; resolves hash
    collisions
    labels jsonb NOT NULL, created_at timestamptz NOT NULL,
    PRIMARY KEY (tenant_id, series_id),
    UNIQUE (tenant_id, name, canonical_labels)
);
CREATE INDEX metric_series_by_name ON metric_series (tenant_id, name, series_id);
CREATE TABLE metric_batches (
    tenant_id text NOT NULL, producer_id text NOT NULL, batch_id text NOT NULL,
    producer_incarnation text NOT NULL, sequence bigint NOT NULL CHECK (sequence >= 0),
    payload_hash text NOT NULL,                                -- same batch_id with a different hash is
    409
    state text NOT NULL CHECK (state IN ('ACCEPTED','PARTIAL','REJECTED','PROCESSED')),
    accepted_count integer NOT NULL CHECK (accepted_count >= 0),
    rejected_count integer NOT NULL CHECK (rejected_count >= 0),
    accepted_at timestamptz NOT NULL, expires_at timestamptz NOT NULL, -- 404 after retention
    PRIMARY KEY (tenant_id, producer_id, batch_id),            -- lost 202: the retry reads the prior
    result
    CHECK (expires_at > accepted_at)
);
CREATE INDEX metric_batches_expiry ON metric_batches (expires_at);
CREATE TABLE rollup_contributions (
    tenant_id text NOT NULL, series_id text NOT NULL,
    resolution_s integer NOT NULL CHECK (resolution_s > 0), window_start timestamptz NOT NULL,
    source_batch_id text NOT NULL,                             -- unique application boundary
    sample_count bigint NOT NULL CHECK (sample_count >= 0),
    sum_value double precision NOT NULL, min_value double precision NOT NULL, max_value double precision NOT
    NULL,
    PRIMARY KEY (tenant_id, series_id, resolution_s, window_start, source_batch_id), -- duplicate insert is a
    no-op
    FOREIGN KEY (tenant_id, series_id) REFERENCES metric_series
);
CREATE TABLE rollup_windows (
    tenant_id text NOT NULL, series_id text NOT NULL,
    resolution_s integer NOT NULL CHECK (resolution_s > 0), window_start timestamptz NOT NULL,
    version bigint NOT NULL CHECK (version > 0),                -- a rebuild replaces a complete version
    atomically
    sample_count bigint NOT NULL CHECK (sample_count >= 0),
    sum_value double precision NOT NULL, min_value double precision NOT NULL, max_value double precision NOT
    NULL,
    histogram jsonb,                                           -- mergeable bucket counts, never
    percentiles
    provisional boolean NOT NULL,                               -- true until the lateness window closes
    PRIMARY KEY (tenant_id, series_id, resolution_s, window_start, version),
    FOREIGN KEY (tenant_id, series_id) REFERENCES metric_series
);
CREATE TABLE metric_replays (
    tenant_id text NOT NULL, replay_id text NOT NULL, idempotency_key text NOT NULL,
    source_from timestamptz NOT NULL, source_to timestamptz NOT NULL,
    transform_version text NOT NULL, actor_id text NOT NULL,
    state text NOT NULL CHECK (state IN ('ACCEPTED','RUNNING','SUCCEEDED','FAILED','CANCELLED')),
    created_at timestamptz NOT NULL, updated_at timestamptz NOT NULL,
    PRIMARY KEY (tenant_id, replay_id),
    UNIQUE (tenant_id, idempotency_key),
    CHECK (source_to > source_from)
);
CREATE INDEX metric_replays_active ON metric_replays (tenant_id, source_from, source_to)
    WHERE state IN ('ACCEPTED','RUNNING');                     -- overlap check behind the 409

```

- Not captured by the DDL
  - Logical points (series, producer incarnation, sample sequence) are TSDB samples partitioned by (tenant, series\_id, time\_bucket); log records (partition, offset) are recovery coordinates that never become rows.

- Consumer progress is the committed log offset; `rollup_contributions` is the correctness boundary that absorbs a redelivered record.
- Tenant series and label budgets are control-plane configuration that gateways cache as snapshots; enforcing them is a count against `metric_series`, not a constraint.
- Deviation from the companion
  - The file's series identity (tenant, canonical name, sorted labels) is the `UNIQUE` constraint; `series_id` stays the primary key so the rollup tables carry a compact foreign key.
  - `metric_batches` gains `producer_incarnation`, `sequence`, `counts`, and `expires_at` for the 202 and 404-after-retention contract; `rollup_contributions`, `rollup_windows`, and `metric_replays` are added for the state table rows and `POST /replays`.
  - The replay overlap check is a partial index plus an application query; a range `EXCLUDE` constraint is the stricter alternative.
- Partition by stable series identity.
  - A consumer may own multiple partitions.
  - Ordinary consumer-group parallelism is bounded by partition count.
  - Splitting one series across partitions abandons simple per-series ordering.
- Batch routing
  - A host batch may contain many series.
  - Gateway can fan records into their series partitions.
  - Partial append failure requires retry-safe point identity.
  - Do not acknowledge a whole batch when only some required records are durable unless the API reports partial acceptance.
  - OCI Monitoring's `PostMetricData` exposes a batch-atomicity choice for this exact case
  - Treat that as documented OCI behavior and the routing above as proposed design.

## Agent concurrency and aggregation — IC3 baseline

```

RECORD(point):
    under short buffer lock:
        validate bounded series count
        apply type-specific aggregation to active buffer

FLUSH():
    under same lock:
        detached = active buffer
        active buffer = new empty buffer
    serialize detached outside lock
    persist/send with stable batch ID
    release retry copy only after durable acceptance
  
```

- Never copy a concurrent map and then remove copied keys as a flush protocol.
  - Writers can update those keys between copy and remove.
  - The removal can discard points that were never included in the sent batch.
- Bound spool bytes and batch age.
  - A row-count limit does not bound memory when labels and payloads vary.
  - Concrete assumption: 64 MiB spool per host and 5 min maximum batch age.
  - Drop low-priority unaccepted telemetry with explicit counters when disk is exhausted.
  - Critical audit or business records require a different contract.
- Type-specific aggregation
  - Counter deltas retain producer reset/incarnation information.
  - Gauges use a declared last-value or min/max sampling policy.
  - Histogram buckets merge only when boundaries are compatible.
  - Average equals total sum divided by total count.

- Averaging independent percentiles does not yield the combined percentile.
- Cardinality
  - Normalize route templates.
  - Reject unbounded user IDs or request IDs as labels.
  - Apply a per-tenant active-series budget.
  - See [Prometheus instrumentation guidance](#).
- Exact failure
  - The lock protects the swap, not the send.
  - A crash after the swap but before durable spooling can lose the detached batch under the stated metrics policy.
  - A dead process cannot reliably increment its own drop counter afterward.
  - Infer the gap from producer sequences or an independent checkpoint, or disclose that exact loss counting is unavailable.

## Delivery, time, and replay — IC3 baseline

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- Producer retries
  - An idempotent broker producer reduces duplicates within its supported scope.
  - Gateway retries and explicit later replay still need stable application identities.
- Sink transaction
  - Insert a unique contribution and update its aggregate in one transaction when supported.
  - On duplicate contribution, do not add again.
  - Commit consumer progress only after the sink state is recoverable.
  - Alternatively materialize deterministic complete windows and replace by version.
- Consistency boundary
  - The log offset is the recovery coordinate
  - The sink's unique contribution key is the correctness boundary.
  - A consumer that commits the offset first and writes second can lose data
  - A consumer that writes first and commits second can duplicate, which the unique key absorbs.
- Event time
  - Keep source observation time and gateway receipt time separate.
  - Accept bounded out-of-order points, for example within 5 min by assumption.
  - Expose provisional rollups until the lateness window closes.
  - Late corrections create a new materialization version or enter a repair workflow.
- Poison versus outage
  - Deterministic schema errors enter quarantine with diagnostic metadata.
  - Sink unavailability should pause/retry with backpressure.
  - Moving every record to DLQ during an outage merely relocates the backlog.
- Kafka transactions do not automatically make an external TSDB exactly once.
  - The external sink must cooperate through idempotency or an atomic progress boundary.
  - See [Kafka delivery semantics](#).

## Failure, operations, and security

| Failure                   | Blast radius                         | Detection                 | Degraded behavior                                          | Recovery                               |
|---------------------------|--------------------------------------|---------------------------|------------------------------------------------------------|----------------------------------------|
| Agent disconnected        | One host                             | Spool age and retry rate  | Bounded local buffering; counted drops when policy permits | Retry stable batches with jitter       |
| Gateway fault domain lost | One fault domain's share of requests | Health checks, error rate | Remaining domains absorb load; agents retry                | Load balancer drains; replace capacity |

| Failure                                                   | Blast radius                         | Detection                       | Degraded behavior                                                              | Recovery                                                      |
|-----------------------------------------------------------|--------------------------------------|---------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------|
| Log quorum unavailable                                    | A partition cohort                   | Produce failures                | Stop durable acknowledgment                                                    | Recover replicas and replay local spool                       |
| TSDB unavailable                                          | Its assigned series                  | Consumer lag age                | Pause consumption; query freshness degrades                                    | Catch up below resource limits                                |
| Series catalog / policy store unavailable (control plane) | New budgets and new-series admission | Snapshot age                    | Gateways accept on cached budgets; unknown new series held to the cached quota | Refresh snapshots; reconcile series created during the outage |
| High-cardinality tenant                                   | Shared index resources               | New-series admission and memory | Reject new series over budget                                                  | Correct instrumentation and expire unused series              |
| Poison record                                             | A bounded record set                 | Classified validation failures  | Quarantine without infinite retry                                              | Replay with stable original identity                          |
| Duplicate replay                                          | Target series/windows                | Unique-key conflicts            | No double aggregate                                                            | Audit replay progress and reconcile counts                    |
| Expensive query                                           | Query/storage resources              | Query cost and queue age        | Cancel or return bounded partial results by contract                           | Tighten fan-out and resolution limits                         |

- Observe
  - Oldest unprocessed record age.
  - Estimated time to retention loss.
  - End-to-end freshness.
  - Spool drops.
  - Quarantine growth.
  - Tenant cardinality admission.
- Retry discipline
  - Agents retry with exponential backoff and jitter under a total elapsed deadline.
  - Retry only transport and 5xx responses
  - A 4xx validation error requires a fix, not a retry.
  - Consumers retry sink writes with backpressure
  - They never skip an offset to make progress.
- Tenant isolation
  - Authenticate producers and bind tenant identity server-side.
  - Apply authorization to labels, queries, and replay operations.
  - Per-tenant budgets cover ingest rate, active series, and query cost.
  - A tenant's query receives 404 on another tenant's series, not 403, to avoid existence leaks.
  - Remove credentials and sensitive payloads from diagnostic logs.
  - Encrypt transport and durable stores.
- Rollout and rollback safety
  - Roll the gateway fleet by fault domain
  - Agents keep spooling, so a bad deploy costs freshness, not accepted data.
  - Consumer schema changes follow expand, migrate, contract
  - A new consumer must read records written by the old gateway.
  - Rollback of a rollout transform is a new materialization version, never an in-place edit.
- Operational drill
  - Stop a sink for a bounded interval.
  - Measure backlog growth and net catch-up rate.
  - Verify replay does not double aggregates.
  - Confirm queries expose stale data instead of silently implying freshness.

## Alternatives and IC4 stretch

| Option                 | What it is                  | Strength                         | Weakness                                     | Choose when                                       |
|------------------------|-----------------------------|----------------------------------|----------------------------------------------|---------------------------------------------------|
| Direct TSDB write      | Synchronous ingestion       | Small topology                   | Sink outages directly affect producers       | Low volume and durable producer buffering         |
| Partitioned replay log | Durable append before sinks | Replay and independent consumers | Retention, partitions, and operating cost    | Baseline sustained shared ingestion               |
| Work queue             | Task delivery               | Simple worker decoupling         | Replay and ordering depend on queue contract | One task consumer without historical reprocessing |
| Raw-only storage       | Preserve all detail         | Flexible future queries          | Storage and read amplification               | Small retention or forensic requirements          |
| Raw plus rollups       | Multiple resolutions        | Lower query/storage cost         | Late correction and versioning complexity    | Repeated time-range dashboards                    |

- IC4 stretch — no accepted loss
  - Rank accepted-data recoverability before query freshness.
  - Remove drop-on-full for accepted records
  - The agent blocks or returns backpressure instead.
  - Reduce raw retention or query precision only with an explicit product agreement.
  - Separate billable metering from lossy operational telemetry.
- IC4 stretch — 10× load
  - Raw daily input becomes approximately 43.2 TB/day before replication.
  - Local aggregation and retention choices can save more than adding brokers.
  - Preserve metric semantics while reducing resolution.
- IC4 stretch — cost and blast radius
  - Isolate large tenants or workload classes into cells.
  - Bound log retention pressure and query blast radius per cell.
  - Migrate series ownership with a routing version and a replay boundary.
  - Replicated log retention is the dominant storage cost
  - Shorten it only after the catch-up drill proves the margin.
- Weakest assumption
  - The sink absorbs 1.5× live rate during catch-up while still serving queries.
  - If the real headroom is 1.1×, a 2 h outage takes 20 h to drain and consumes nearly the whole 24 h retention window.
  - Mitigation: measure catch-up under query load, shed rollup work during catch-up, and alarm on estimated time to retention loss.
- Over-engineering warning
  - Do not call the pipeline “zero loss” without naming the acknowledgment boundary and surviving failure set.
  - Do not add Kafka transactions expecting them to make the external database exactly once.
- Related business prompt
  - [Healthcare ingestion](#) has stricter record retention and reconciliation requirements.
  - Its accepted business records must not inherit the metrics drop policy.

## Level signal — IC3 baseline vs IC4 stretch

| Phase        | IC3 must land                                                    | IC4 adds                                                                   | Separating signal                                   |
|--------------|------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------|
| Requirements | Metric types, loss policy, freshness target, tenant in every key | Separates billing-grade from telemetry and names what the drop policy buys | Negotiates the loss contract instead of assuming it |
| Estimates    | One million points/s, backlog and catch-up math, cardinality     | Lets catch-up headroom set retention; names the correlated-outage case     | The catch-up number changes the retention decision  |

| Phase        | IC3 must land                                                     | IC4 adds                                                                     | Separating signal                               |
|--------------|-------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------|
| Architecture | Gateway, log, consumers, query tier with one trace                | Control-plane snapshots so ingestion survives a catalog outage               | Every component has a mechanism and a price     |
| Deep dive    | Swap-buffer flush, stable point identity, unique contribution key | Offset-first versus write-first consumer failure analysis; versioned rollups | Reasons across the log and sink boundary        |
| Operations   | Failure table, freshness alarms, outage drill                     | Cells, fault-domain rollout, retention as a cost control                     | Volunteers the drill before being asked         |
| Wrap         | Recap and tradeoff                                                | Names the catch-up assumption as the weakest first                           | Identifies the flaw before the interviewer does |

## Interview pushback and validation

- Q1: Does batching reduce one million points to ten thousand points?
  - **IC3 baseline**
    - No
    - This example reduces HTTP requests to ten thousand per second.
  - **IC4 stretch**
    - Only semantically valid aggregation can reduce logical observations.
- Q2: A consumer crashes after writing but before committing its offset.
  - **IC3 baseline**
    - It reprocesses
    - A unique sink identity prevents double application.
  - **IC4 stretch**
    - An aggregate needs atomic dedup-plus-update, not a separate best-effort dedup cache.
- Q3: Can we average host p99 latency values?
  - **IC3 baseline**
    - No
    - Merge distributions or compatible histogram buckets.
  - **IC4 stretch**
    - Choose resolution and error budget before comparing histogram formats.
- Q4: The data is now billing-grade with no accepted loss.
  - **IC3 baseline**
    - Remove drop-on-full for accepted records and enforce durable acceptance/backpressure.
  - **IC4 stretch**
    - Design reconciliation and regional durability separately from dashboard telemetry.
- Q5: Is one partition per consumer mandatory?
  - **IC3 baseline**
    - A consumer can own several partitions
    - One partition has one active group owner.
  - **IC4 stretch**
    - Partition reassignment must coordinate progress and sink effects.
- Q6: Why is query deployment isolation insufficient?
  - **IC3 baseline**
    - Ingestion and queries may still compete for the same storage resources.
  - **IC4 stretch**
    - Reserve resources or isolate read replicas/materialized blocks.
- Q7 (premise reversal): Why not drop the log and write directly to the TSDB with a bigger agent spool?
  - **IC3 baseline**
    - A sink outage then becomes 100,000 independent recoveries, replay has no shared ordered source, and the agent spool is the weakest disk in the fleet.

- **IC4 stretch**
  - Acceptable only for low volume with a durable producer store
  - At one million points/s the log's cost is cheaper than fleet-wide spool operations.
- Q8: The series catalog is down
  - Do we stop accepting points?
  - **IC3 baseline**
    - No
    - Gateways accept on cached budgets and reconcile new series after the catalog returns.
  - **IC4 stretch**
    - Bound the reconciliation debt with a cached quota for unknown series so the outage cannot become a cardinality bypass.
- Validation scenarios
  - Agent record racing with flush.
  - Partial batch append and retry.
  - Sink write followed by consumer crash.
  - Replay of an existing aggregate contribution.
  - Producer counter reset.
  - Late samples after a rollup closes.
  - Recovery before log retention expires.
  - Catalog outage while a tenant creates new series.

## Recall card

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- "The agent batches without losing concurrent writes."
- "Durable append separates acceptance from query freshness."
- "Stable point identity and an atomic sink boundary make replay safe."
- "Cardinality, retention, and catch-up capacity are first-class budgets."
- "The catalog is control plane; acceptance must not depend on it."
- Wrap-up script
  - "The design is a swap-buffer agent, a gateway that acknowledges after a durable log quorum, and idempotent consumers feeding a TSDB with rollups."
  - "An acknowledgment marks durable log acceptance, and sink-side contribution identity protects replay only when stored atomically with aggregation."
  - "The main tradeoff is the log itself: it buys replay and outage isolation at the price of retention bytes and a second system to run."
  - "My weakest assumption is 1.5× catch-up headroom; I would run the sink-outage drill first to validate it."

## References

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- [Ranked question and report provenance.](#)
- [OCI role framing.](#)
- [Detailed API and schema companion.](#)
- Reference check: 2026-09-09.
  - Official documentation supports the mechanisms cited next to each link.
  - Capacity values and architecture choices remain proposed interview assumptions.
- [Kafka design](#): partitioned delivery and the external-sink boundary.
- [Prometheus instrumentation](#): cardinality and metric semantics.
- [Prometheus metric types](#): counters, gauges, and distribution metrics.
- [OCI Monitoring PostMetricData](#): batch atomicity option for partial acceptance
  - Documented OCI behavior, not this design.